

**Problem # 1:** Multiple choice question: [10 x 1pt]

1	2	3	4	5	6	7	8	9	10
B (5)	B	D	A	D	B	A	A	A	A

1. Find the content of AL and ZF register after execution of the following code:

```
MOV AL,FCH
TEST AL,16
```

- A. AL=FCH; ZF=1 **B. AL=FCH; ZF=0** C. AL=00H; ZF=1 D. AL=0CH; ZF=0

$$\begin{array}{r} 1111 \ 1100 \\ 0001 \ 0000 \\ \hline 0000 \ 0000 \\ \text{ZF}=1 \end{array}$$

2. Instruction pointer (IP) contains the offset address of _____ segment.

- A. Data segment **B. Code segment** C. Stack segment D. Extra segment

3. The output of the following operation on register BL is:

```
MOV BL,FCH
MOV CL,08
SHL BL,CL
```

- A. BL= CFH** B. BL= FCH C. BL= OFH D. BL=00

$$\begin{array}{r} 1111 \ 1100 \\ 0000 \ 1000 \\ \hline 1100 \ 0000 \end{array} \ll 8$$

4. In the following instruction sequence, the values of the parity, auxiliary carry, and sign flags are:

```
MOV AX, 0C17FH
ADD AX, 3
```

- A. PF=0, AF=1, SF=0 B. PF=1, AF=1, SF=1 C. PF=0, AF=0, SF=1 **D. PF=1, AF=0, SF=0**

$$\begin{array}{r} 0C17F \\ + 00003 \\ \hline 0C182 \end{array} \quad \begin{array}{l} \text{PF}=1, \text{ AF}=0, \\ \text{SF}=0 \end{array}$$

5. If the offset address of X in data segment is equal to 230H, and DS=2500H, then the physical address of X is?

- A. 20315 H **B. 25320H** C. 2500 H D. 323000

$$25000 + 320 = 25320$$

6. Which of the following is an example of a relative based addressing mode?

- A. MOV AX, [BP + SI + 10h]** B. MOV AX, [BP+1] C. MOV AX, [SI+1] D. MOV AX, [BX]

7. Suppose X defined as below; what is the content of CX, and AH registers after execution of the following instructions:

```
X    DW 4580H,0ABCH,1100,2222H
MOV BX, OFFSET X
MOV AH,1
MOV CX,[BX]
INC BX
ADD AH, [BX]
```

X	BX
+0	4580
+2	0ABC
+4	1100
+6	2222

$$CX = 4580H$$

$$AH = 01 + 0ABCH = 0ABDH$$

- A. CX=4580H; AH=46H** B. CX=0ABCH; AH=0AH
C. CX=BCD0H; AH=B0H D. CX=BC45H; AH=0CH

8. Assuming DS=4000h, SI=10h, and BX=23h, the physical address of the source operand in the instruction MOV AX, [BX + SI + 0C3h] is,

- A. 400F6h** B. 4FFF6h C. 40033h D. 40123h

$$10 + 23 + C3 = F6 \\ 40000 + F6 = 400F6$$

9. The value of AX at the end of execution of the following 8086 assembly instruction sequence is:

```
MOV AL, -9
SHL AX, 4
```

- A. AL=70h** B. AL=0Fh C. AL= 60h D. AL= 80h

$$-9 = F7 \text{ (in AL)} \\ AX = 00F7 \\ SHL 4 \rightarrow 0F70$$

10. In the following instruction sequence, show the value of AL:

```
MOV AL,0FFH
NEG AL
XOR AL,FFH
```

- A. AL= FEH** B. AL= F15H C. AL= 0FH D. AL= 0FFH

$$AL = FF \\ NEG \rightarrow 01 \\ XOR FF \rightarrow FE$$

**Problem # 2:** Find the contents of the registers written below for each of the following assembly codes: [10 pts]

A)

```

MOV AX,0FFE0H
MOV CX,0000
PUSH AX
PUSH BX
POP DX
MOV X,DX
POP CX
HLT

```

0010
000E
??
FF

AX = FFE0H CX = FFE0 DX = 0000
 BX = 0000H SI = 0000H

B)

```

X DB 1,2,3,4,5,6,7,9
MOV CX,8
MOV SI,0
L2: CMP X[SI],6
JB L
ADD AL,X[SI]
L: INC SI
CMP SI,4

```

AL = 06H SI = 00H

C)

```

X DW 35FAH, 48CBH, 231AH, 0080H
MOV BX, OFFSET X
ADD BX,2
MOV AL,[BX]
INC BX
MOV DX,[BX]

```

FA	35
31	37
CB	39
48	3B
1A	3D
25	3F
80	41

AL = CBH DX = 1A48
 BX = 0014H

D)

```

MOV AL,FEH
ADD AL,2
ADC AL,3

```

```

  1111 1110
+ 0000 0010
-----
  1 0000 0000
    CF = 1
    + 0000 0011
    -----
    0000 0100

```

AL = 04H ZF = 1

Problem # 3:A) Find the result of DX without using multiplication and division instructions $DX = 5*AX + 9*BX$

```

MOV CX, AX
MOV DX, BX
SHL AX, 2
SHL AX, 2
ADD AX, AX
MOV CL, 3
SHL BX, CL

```

```

MOV DX, AX
ADD DX, BX

```

$$\begin{aligned}
 DX &= 5*AX + 9*BX \\
 DX &= 4AX + 1AX + 8BX + BX \\
 &= 5AX + 9BX
 \end{aligned}$$

B) Write the equivalent of the following instructions

- SUB AX, BX $\rightarrow$ { ① MOV DX, AX
NEG BX
ADD AX, BX
NEG BX } 1
- XCHG AX, BX $\rightarrow$ { ② MOV DX, AX
MOV AX, BX
MOV BX, DX } 2
- NOT AX $\rightarrow$ { ③ XOR AX, FFFF } 3